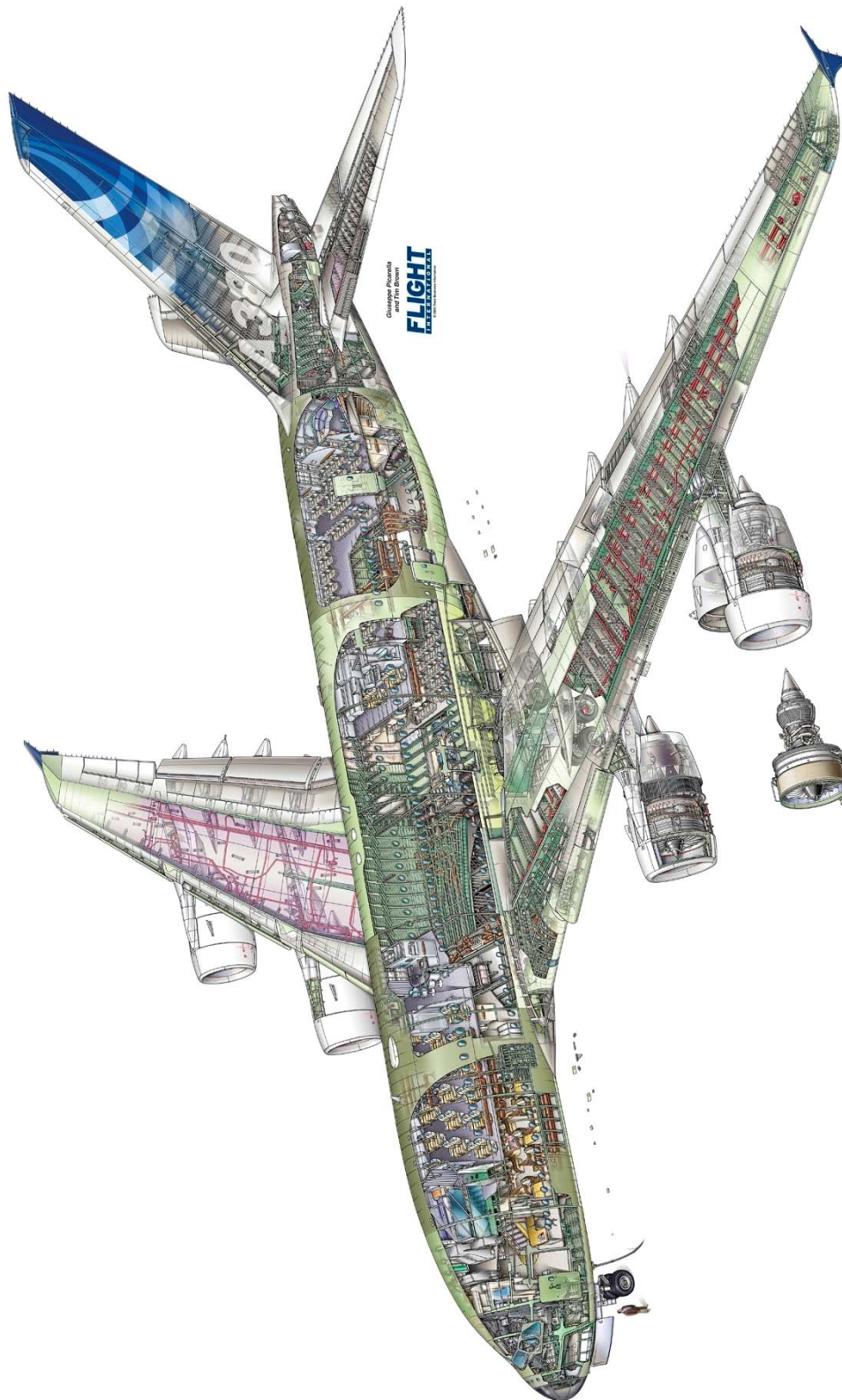


AIRCRAFT STRUCTURAL ELEMENTS, LOADS AND MATERIALS



Book List

Aircraft Structures for Engineering Students
Megson. Arnold press.

Aircraft Loading and Structural Layout
D.Howe Professional Engineering Publishing

Understanding Aircraft Structures
John Cutler. BSP professional books.

Analysis and Design of Flight Vehicle Structures
Bruhn.

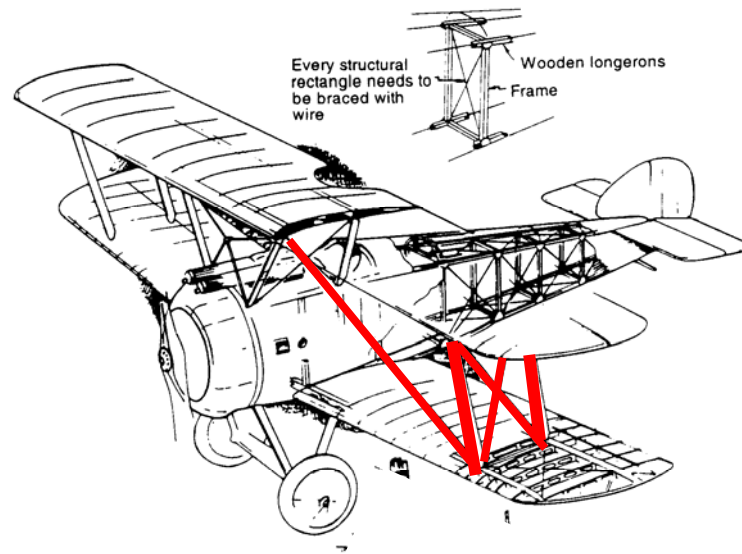
Airframe Structural Design.
M.C.Y. Niu. Conmilit press ltd.

Engineering Sciences Data Unit ESDU.

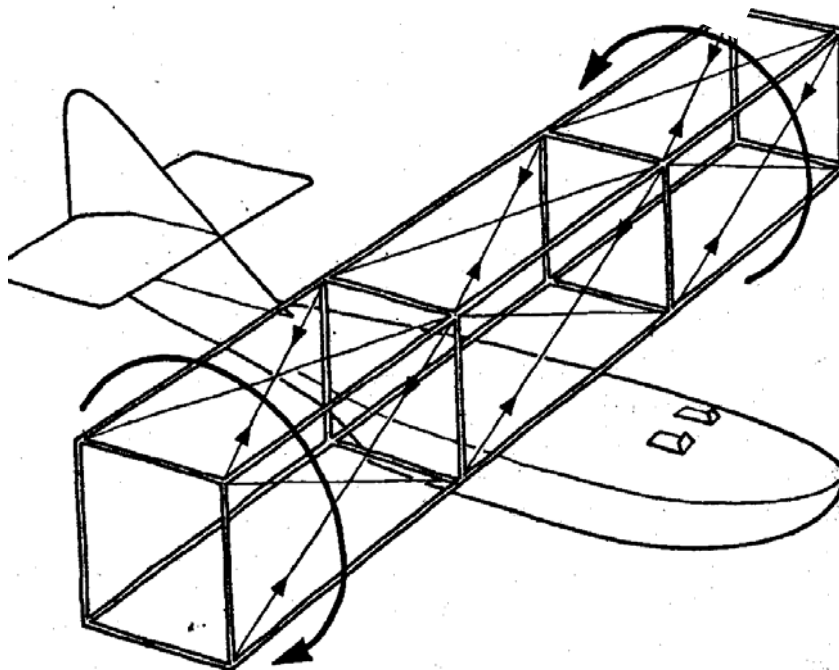
Stresses in Aircraft and Shell Structures
P.Kuhn McGraw-Hill

Mechanics of Materials.
Beer and Johnson. McGraw-Hill.

Formulas for Stress and Strain.
Roark. McGraw-Hill.



Sopwith Camel (1917)



INTRODUCTION

The basic function of an aircraft's structure is to resist and transmit the applied loads with minimum weight, to provide an aerodynamic shape and to protect the systems and passengers from the environmental conditions of the flight. These requirements basically result in thin shell structures where the outer skin is supported by frames and stiffeners to enable it to withstand compressive bending and torsional loads without buckling or fracturing.

Such structures are known as semi-monocoque. Thin shells which rely entirely on their skins to resist loads are referred to as monocoque.

Unfortunately no structure can be made perfect, as access doors, panels and windows are added in order that the structure may fulfil its final role. These introduce discontinuities in the structure around which the loads will have to pass. In the final design reinforcing will have had to be added at many of these features, each having required careful analysis.

In the next five topics we will consider how the different types of structural element are used in stressed skin construction, and the form of internal loading that arises for given external loads.

WING STRUCTURE

A typical wing structure is shown in Figure 1. Three major structural elements are used in the wing, namely ribs, skin and spar beams.

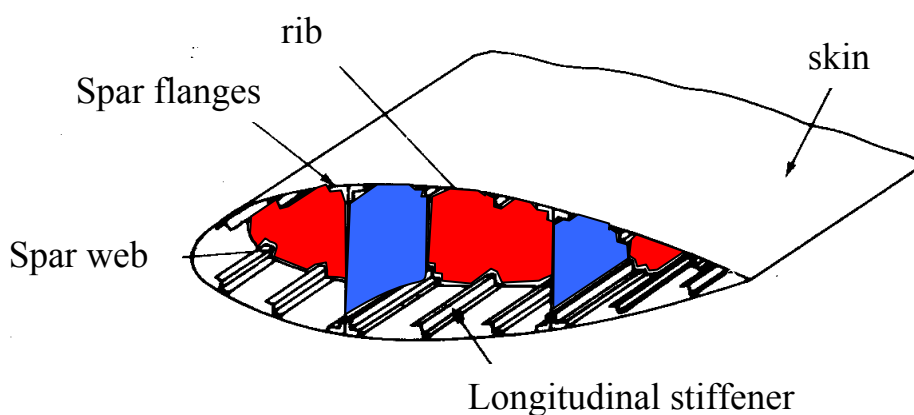


Figure 1

Ribs

The shape of the wing is governed by aerodynamic considerations and clearly must be maintained for all load combinations. This is just one of the functions of the ribs or transverse frames. The ribs are also used to distribute concentrated loads (such as undercarriage, engine or wing store loads) into the structure. They also help to reduce the buckling length of the longitudinal stiffeners by providing end restraints, thus increasing the buckling load which can be carried by the stiffeners. They provide attachment points for flaps and other control surfaces and obviously vary in strength depending upon the loads they are required to carry. The heavier ribs are often referred to as bulkheads. A typical rib construction is shown in Figure 2.

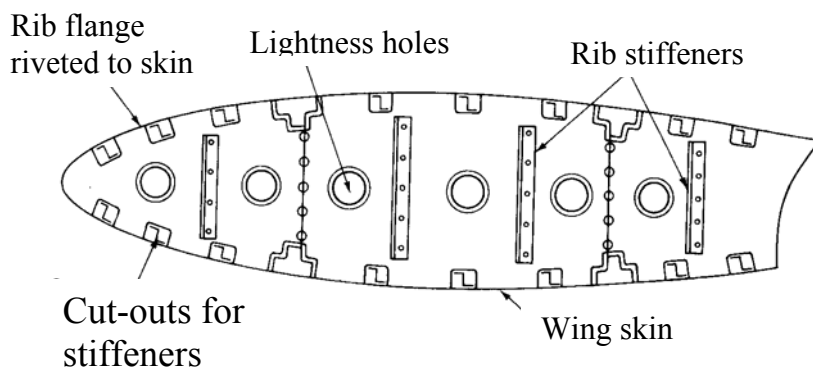


Figure 2

Skin

The prime function of the wing skin is to form an impermeable surface for supporting the aerodynamic pressure distribution from which the lifting capability of the wing is derived.

These aerodynamic forces are transmitted in turn to the ribs and stiffeners by the skin through plate and membrane action. Resistance to shear and torsional loads is supplied by shear stresses developed in the skin and spar webs while axial and bending loads are reacted by the combined action of skin and longitudinal stiffeners.

Now the thin skin is efficient for resisting shear and tensile loads but it buckles under relatively low compression loads. Instead of increasing the skin thickness however, a more mass efficient solution is to attach longitudinal stiffeners to the skin thereby dividing the skin into small panels and increasing the buckling load of the skin. This effect also works the other way with the skin increasing the buckling load of the stiffeners.

Spar Beams

The spar webs develop shear stresses to resist shear and torsional loads as already mentioned. They also perform a secondary function in stabilising the skin and help in resisting axial and bending loads.

FUSELAGE

The fuselage structure is composed of structural members which perform similar functions to their equivalents in the wing. The loads that the fuselage has to carry do differ from those of the wing. For instance, the aerodynamic loads on the fuselage are low but the fuselage is pressurised. The shape of the fuselage cross-section is basically dictated by having to carry the pressure load, a circular cross section being the most efficient, but deviations from this may be dictated by operational requirements. The fuselage also has to carry large concentrated loads such as the wing and tail loads. Figure 3 shows a simple representation of the fuselage structural elements.

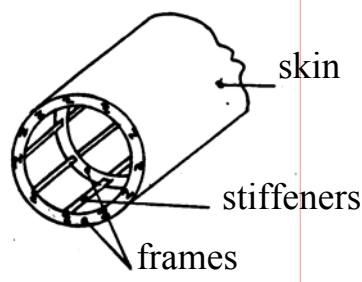


Figure 3

There are three major fuselage elements namely skin, frames and stiffeners. As for the wing, the combination of stiffeners and frames reduce the skin to small areas which increases the skins resistance to buckling. The frames constrain the stiffeners to short lengths thus increasing their buckling strength. The frames also help to maintain the required shape of the fuselage cross section doing a job equivalent to the wing ribs. The fuselage supports large concentrated loads such as the wings and undercarriage, and also must withstand pressure differential loads. Cut-outs in the skin for such items as windows, undercarriage etc., a framed reinforcement is used plus possible local doubling of the skin and the addition of stringers.

The ideal flight structure would be one single complete unit of the same material involving one manufacturing operation. This, however, is not possible for many reasons such as mass, maintenance and manufacturing methods. Joints are basically an inefficient load path and thus add mass to the structure. The number of joints in a lightweight structure should, therefore, be minimised and their locations should be optimised and not located in high load paths.

TYPES OF PANEL

This section looks at the main types of lightweight panel options for light weight structures.

a) Sandwich Construction Panels

Sandwich panel construction consists of faceskins, core and core to faceskin bonding material. A sandwich panel is equivalent to an 'I' beam with the facings corresponding to the flanges and the core corresponding to the web of an 'I' beam. The facing skins carry axial tensile and compressive loads while the core carries shear loads and stresses and compressive stresses normal to the panel. The core supports the face skins and resists buckling of the skins. Typical facing skin materials are aluminium alloy, titanium alloy, carbon fibre reinforced plastic and glass FRP. Figure 4 shows details of panel construction.

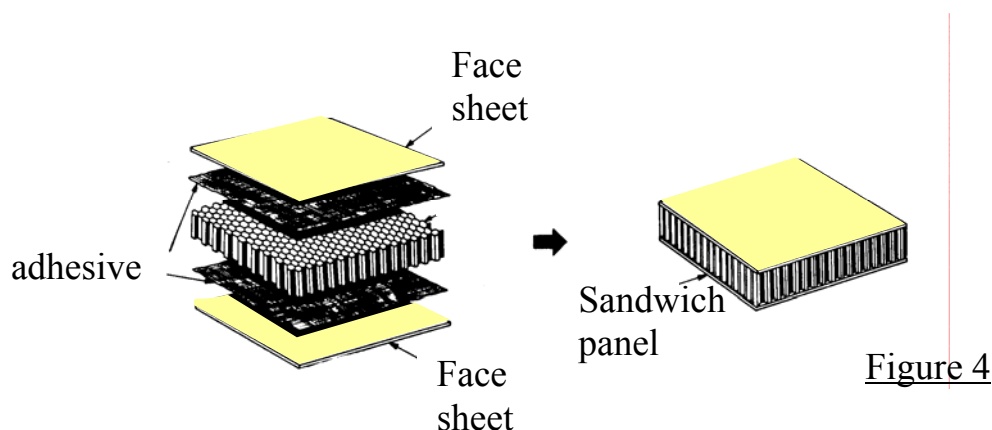


Figure 4

Typical core materials are:-

Metallic cores – aluminium alloy, titanium alloy and stainless steel. These materials give high shear strength and shear modulus. Aluminium is generally preferred for low cost but for high temperature applications titanium or stainless steel may be used.

Non-metallic cores –common types are foams, balsa wood, FRPs in the form of Honeycomb.

Honeycomb panels are widely used in aircraft to-day in areas such as ailerons, trailing edges, access doors, dive brakes, flooring, bulkheads, fuselage sections, elevators, rudders etc.

b) Integral Skin - Stiffener Panels

In an integral skin-stiffener panel, the skin and the stiffeners are machined as one from thick plate by numerically controlled machines. Something in the order of 80% of the plates volume maybe machined away but the finished panel can still be cost competitive with a riveted skin stringer panel but the integral panel is technically superior in a number of ways –for instance stress concentrations at rivet holes are eliminated, also fretting between skin and stiffeners. Furthermore, the thickness of a machined skin can be varied to suit the local load levels (see Figure 5) Even so, joints are not completely eliminated as the integral panels have to be riveted to the hoop frames and to each other. The cheaper the material and energy, the more competitive these panels become.

The majority of the Concorde fuselage panels are integrally machined. They are also used in the wing structure.

In summary therefore, sandwich panels, semi-monoque and integrally machined panels are all technically competitive and are favoured depending upon the application and prevailing capabilities and economic circumstances.

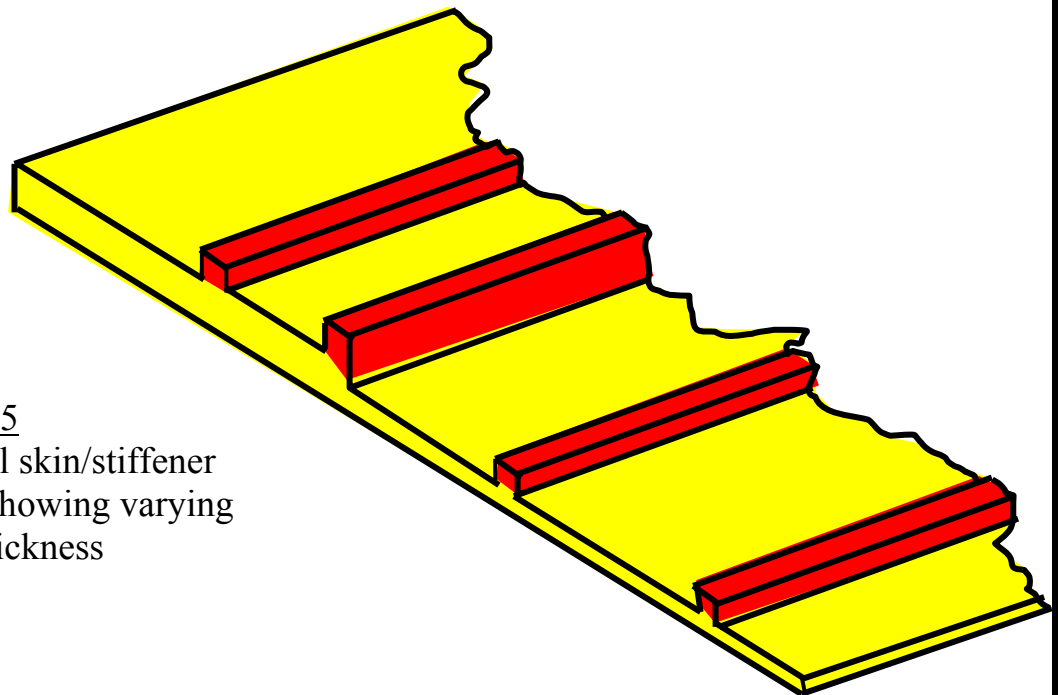


Figure 5
Integral skin/stiffener
panel showing varying
skin thickness

REQUIREMENTS AND LOADS

Regulatory authorities exist within the civil aircraft industry, three main design standards need to be considered; for military aircraft each countries defence agency (MOD) will control the regulations:-

- BCAR British civil airworthiness requirements (basically superseded)
- JAR Joint airworthiness requirements
- FAR Federal aviation regulations

The three publications cover British, European and American authorities. However, for all practical purposes the JAR covers the contents of the other two, although care needs to be exercised. These are legal documents and the requirements have to be adhered to and demonstrated to the relative authority during the aircraft certification process.

A typical contents section of the JAR is given on the following page, JAR-25 large aircraft, together with an extract of the structure section covering loading envelopes.

- Flight Envelopes

Included in the JAR are loading curves (flight envelopes) that from experience have been shown to provide load factors that provide an acceptable safe design, e.g. for a civil aircraft the probability of exceeding 2.5g during a manoeuvre has proven to be acceptably low.

The strength requirements of the aircraft must be met at all points of the flight envelope. The manoeuvre envelope gives a maximum of 2.5g and a minimum of -1g for a civil airliner. Loads of the magnitude prescribed by the flight envelope are considered to be experienced once in an aircraft life and may cause some yield to the aircraft structure. The forces described in the manoeuvre envelope are largely within the pilots control, a second set of factors prescribed by the gust envelope represent the loading imposed on the structure by external (atmospheric) forces.

The airframe structure will have to be considered for loads from each of the envelopes. So far the effect of the airborne flight phase has been considered but ground loading and fatigue loading has to be added to this for a complete picture to emerge.

SUBPART C — STRUCTURE

GENERAL

JAR 25.301 Loads

(a) Strength requirements are specified in terms of limit loads (the maximum loads to be expected in service) and ultimate loads (limit loads multiplied by prescribed factors of safety). Unless otherwise provided, prescribed loads are limit loads.

(b) Unless otherwise provided the specified air, ground, and water loads must be placed in equilibrium with inertia forces, considering each item of mass in the aeroplane. These loads must be distributed to conservatively approximate or closely represent actual conditions. Methods used to determine load intensities and distribution must be validated by flight load measurement unless the methods used for determining those loading conditions are shown to be reliable. (See ACJ 25.301 (b).)

(c) If deflections under load would significantly change the distribution of external or internal loads, this redistribution must be taken into account.

JAR 25.303 Factor of safety

Unless otherwise specified, a factor of safety of 1.5 must be applied to the prescribed limit load which are considered external loads on the structure. When a loading condition is prescribed in terms of ultimate loads, a factor of safety need not be applied unless otherwise specified.

JAR 25.305 Strength and deformation

(a) The structure must be able to support limit loads without detrimental permanent deformation. At any load up to limit loads, the deformation may not interfere with safe operation.

(b) The structure must be able to support ultimate loads without failure for at least 3 seconds. However, when proof of strength is shown by dynamic tests simulating actual load conditions, the 3-second limit does not apply. Static tests conducted to ultimate load must include the ultimate deflections and ultimate deformation induced by the loading. When analytical methods are used to show compliance with the ultimate load strength requirements, it must be shown that —

- (1) The effects of deformation are not significant;
- (2) The deformations involved are fully accounted for in the analysis; or
- (3) The methods and assumptions used are sufficient to cover the effects of these deformations.

JAR 25.305 (continued)

(c) Where structural flexibility is such that any rate of load application likely to occur in the operating conditions might produce transient stresses appreciably higher than those corresponding to static loads, the effects of this rate of application must be considered.

(d) The dynamic response of the aeroplane to vertical and lateral continuous turbulence must be taken into account. (See ACJ 25.305 (d).)

JAR 25.307 Proof of structure

(a) Compliance with the strength and deformation requirements of this Subpart must be shown for each critical loading condition. Structural analysis may be used only if the structure conforms to those for which experience has shown this method to be reliable. In other cases, substantiating load tests must be made. Where substantiating load tests are made these must cover loads up to the ultimate load, unless it is agreed with the Authority that in the circumstances of the case, equivalent substantiation can be obtained from tests to agreed lower levels. (See ACJ 25.307.)

(b) Compliance with the fatigue evaluation requirements of JAR 25.571 and 25.573 must be shown.

(c) Certain parts of the structure must be tested as specified in JAR 25.601.

(d) When static or dynamic tests are used to show compliance with the requirements of JAR 25.305 (b) for flight structures, appropriate material correction factors must be applied to the test results, unless the structure, or part thereof, being tested has features such that a number of elements contribute to the total strength of the structure and the failure of one element results in the redistribution of the load through alternate load paths.

FLIGHT LOADS

JAR 25.321 General

(a) Flight load factors represent the ratio of the aerodynamic force component (acting normal to the assumed longitudinal axis of the aeroplane) to the weight of the aeroplane. A positive load factor is one in which the aerodynamic force acts upward with respect to the aeroplane.

JAR 25.321 (continued)

(b) Considering compressibility effects at each speed, compliance with the flight load requirements of this Subpart must be shown —

- (1) At each critical altitude within the range of altitudes selected by the applicant;
- (2) At each weight from the design minimum weight to the design maximum weight appropriate to each particular flight load condition; and
- (3) For each required altitude and weight, for any practicable distribution of disposable load within the operating limitations recorded in the Aeroplane Flight Manual.

**FLIGHT MANOEUVRE
AND GUST CONDITIONS**

JAR 25.331 General

(a) *Procedure.* The analysis of symmetrical flight must include at least the conditions specified in sub-paragraphs (b) to (d) of this paragraph. The following procedure must be used:

- (1) Enough points on the manoeuvring and gust envelopes must be investigated to ensure that the maximum load for each part of the aeroplane structure is obtained. A conservative combined envelope may be used.
- (2) The significant forces acting on the aeroplane must be placed in equilibrium in a rational or conservative manner. The linear inertia forces must be considered in equilibrium with thrust and all aerodynamic loads, while the angular (pitching) inertia forces must be considered in equilibrium with thrust and all aerodynamic moments, including moments due to loads on components such as tail surfaces and nacelles. Critical thrust values in the range from zero to maximum continuous thrust must be considered.
- (3) Where sudden displacement of a control is specified, the assumed rate of control surface displacement may not be less than the rate that could be applied by the pilot through the control system.
- (4) In determining elevator angles and chordwise load distribution (in the manoeuvring conditions of sub-paragraphs (b) and (c) of this paragraph) in turns and pullups, the effect of corresponding pitching velocities must be taken into account. The in-trim and out-of-trim flight conditions specified in JAR 25.255 must be considered.

JAR 25.331 (continued)

(b) *Manoeuvring balanced conditions.* Assuming the aeroplane to be in equilibrium with zero pitching acceleration, the manoeuvring conditions A through I on the manoeuvring envelope in JAR 25.333 (b) must be investigated.

(c) *Manoeuvring pitching conditions.*

(1) *Maximum elevator displacement at V_A .* The aeroplane is assumed to be flying in steady level flight (point A1, JAR 25.333 (b)) and, except as limited by pilot effort in accordance with JAR 25.397 (b), the pitching control is suddenly moved to obtain extreme positive pitching acceleration (nose up). In defining the tail load condition, the response of the aeroplane may be taken into account. Loads occurring beyond the point in time where normal acceleration at the c.g. exceeds the maximum positive limit manoeuvring factor n may be ignored.

(2) *Checked manoeuvre between V_A and V_D .* A checked manoeuvre, based on a rational pitching control motion versus time profile must be established in which the design limit load factor specified in JAR 25.337 will not be exceeded. (See also ACJ 25.331 (c) (2).)

(d) *Gust conditions.* The gust conditions B' to J', JAR 25.333 (c), must be investigated. The following provisions apply:

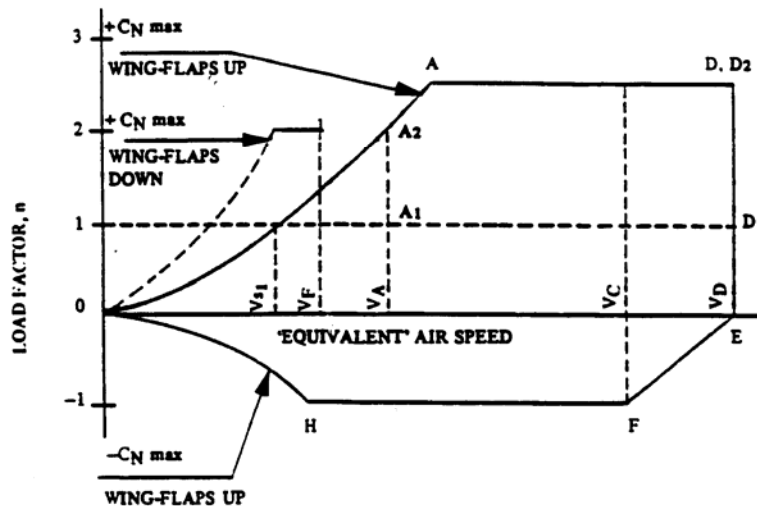
- (1) The air load increment due to a specified gust must be added to the initial balancing tail load corresponding to steady level flight.
- (2) The alleviating effect of wing downwash and of the aeroplane's motion in response to the gust may be included in computing the tail gust load increment.
- (3) Instead of a rational investigation of the aeroplane response, the gust alleviation factor K_g may be applied to the specified gust intensity for the horizontal tail.

JAR 25.333 Flight envelope

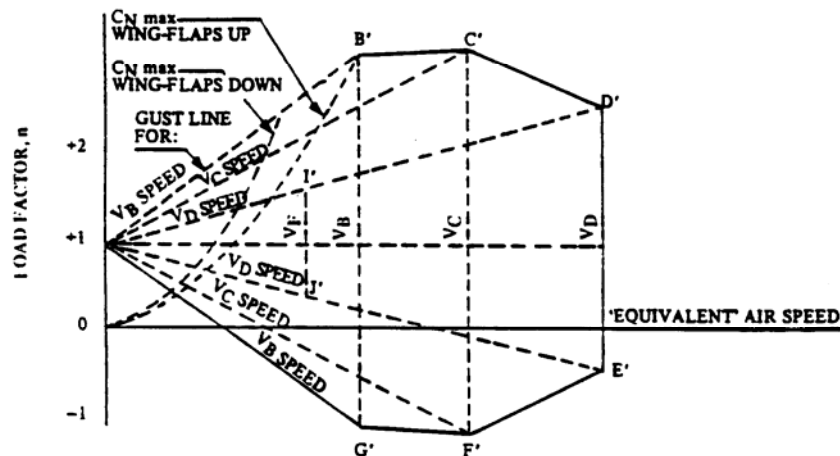
(a) *General.* The strength requirements must be met at each combination of airspeed and load factor on and within the boundaries of the representative manoeuvring and gust envelopes (V - n diagrams) of sub-paragraphs (b) and (c) of this paragraph. These envelopes must also be used in determining the aeroplane structural operating limitations as specified in JAR 25.1501.

JAR 25.333 (continued)

(b) *Manoeuvring envelope.*



(c) *Gust envelope.*



JAR 25.335 Design airspeeds

The selected design airspeeds are equivalent airspeeds (EAS). Estimated values of V_{S0} and V_{S1} must be conservative.

(a) *Design cruising speed, V_c .* For V_c , the following apply:

(1) The minimum value of V_c must be sufficiently greater than V_b to provide for inadvertent speed increases likely to occur as a result of severe atmospheric turbulence.

(2) In the absence of a rational investigation substantiating the use of other values, V_c may not be less than $V_{Bmin} + 43$ knots. However, it need not exceed the maximum speed in level flight at maximum continuous power for the corresponding altitude.

(3) At altitudes where V_d is limited by Mach number, V_c may be limited to a selected Mach number. (See JAR 25.1505.)

JAR 25.335 (continued)

(b) *Design dive speed, V_D .* V_D must be selected so that V_C/M_C is not greater than 0.8 V_D/M_D , or so that the minimum speed margin between V_C/M_C and V_D/M_D is the greater of the following values:

(1) From an initial condition of stabilised flight at V_C/M_C , the aeroplane is upset, flown for 20 seconds along a flight path 7.5° below the initial path, and then pulled up at a load factor of 1.5 g (0.5 g acceleration increment). The speed increase occurring in this manoeuvre may be calculated if reliable or conservative aerodynamic data issued. Power as specified in JAR 25.175 (b) (1) (iv) is assumed until the pullup is initiated, at which time power reduction and the use of pilot controlled drag devices may be assumed;

(2) The minimum speed margin must be enough to provide for atmospheric variations (such as horizontal gusts, and penetration of jet streams and cold fronts) and for instrument errors and airframe production variations. These factors may be considered on a probability basis. However, the margin at altitude where M_C is limited by compressibility effects may not be less than 0.05 M. (See ACJ 25.335 (b) (2).)

(c) *Design manoeuvring speed, V_A .* For V_A , the following apply:

(1) V_A may not be less than $V_{S1} \sqrt{n}$ where —

(i) n is the limit positive manoeuvring load factor at V_C ; and

(ii) V_{S1} is the stalling speed with wing-flaps retracted.

(2) V_A and V_S must be evaluated at the design weight and altitude under consideration.

(3) V_A need not be more than V_C or the speed at which the positive $C_{N_{max}}$ curve intersects the positive manoeuvre load factor line, whichever is less.

(d) *Design speed for maximum gust intensity, V_B*

(1) V_B may not be less than the speed V_{Bmin} , determined by the intersection of the line representing the maximum position lift $C_{N_{max}}$ and the line representing the rough air gust velocity on the gust $V-n$ diagram, or $(\sqrt{n_g}) V_{S1}$, whichever is less, where —

(i) n_g is the positive aeroplane gust load factor due to gust at speed V_C (in accordance with JAR 25.341) and at the particular weight under consideration; and

(ii) V_{S1} is the stalling speed with the flaps retracted at the particular weight under consideration.

JAR 25.335 (d) (continued)

(2) V_B need not be greater than V_C . (See JAR 25X1517 and JAR 25.1585.)

(e) *Design wing-flap speeds, V_F .* For V_F , the following apply:

(1) The design wing-flap speed for each wing-flap position (established in accordance with JAR 25.697 (a)) must be sufficiently greater than the operating speed recommended for the corresponding stage of flight (including balked landings) to allow for probable variations in control of airspeed and for transition from one wing-flap position to another.

(2) If an automatic wing-flap positioning or load limiting device is used, the speeds and corresponding wing-flap positions programmed or allowed by the device may be used.

(3) V_F may not be less than —

(i) 1.6 V_{S1} with the wing-flaps in take-off position at maximum take-off weight;

(ii) 1.8 V_{S1} with the wing-flaps in approach position at maximum landing weight; and

(iii) 1.8 V_{S0} with the wing-flaps in landing position at maximum landing weight.

(f) *Design drag device speeds, V_{DD} .* The selected design speed for each drag device must be sufficiently greater than the speed recommended for the operation of the device to allow for probable variations in speed control. For drag devices intended for use in high speed descents, V_{DD} may not be less than V_D . When an automatic drag device positioning or load limiting means is used, the speeds and corresponding drag device positions programmed or allowed by the automatic means must be used for design.

JAR 25.337 Limit manoeuvring load factors

(a) Except where limited by maximum (static) lift coefficients, the aeroplane is assumed to be subjected to symmetrical manoeuvres resulting in the limit manoeuvring load factors prescribed in this paragraph. Pitching velocities appropriate to the corresponding pull-up and steady turn manoeuvres must be taken into account.

(b) The positive limit manoeuvring load factor 'n' for any speed up to V_D may not be less than $2.1 + \left(\frac{24\,000}{W + 10\,000} \right)$ except that 'n' may not be less than 2.5 and need not be greater than 3.8 — where 'W' is the design maximum take-off weight (lb).

Ground Loads

Taxi-ing and Lift-off: maximum engine thrust and runway loads

Landing: concentrated shock loads.

Handling loads: engine replacement, towing etc.

AIRCRAFT MATERIALS

The following is a typical material data sheet.

		SECTION 6 ALUMINIUM ALLOYS										2L 97									
		PLATE T351										PREVIOUSLY: DTD 5090									
PHYSICAL PROPERTIES																	PRINCIPAL ALLOYING ELEMENTS(%)				
Temperature °C	at 20	at 100	20-100	20-200	20-300	Element												Minimum	Maximum		
$\alpha \times 10^6 / ^\circ\text{C}$			23.3	23.9	24.7	Cu												3.8	4.9		
k W/m K		125				Mg												1.2	1.8		
ρ kg/m ³	2.77					Mn												0.30	0.9		
σ MS/m	17.4																				
c kJ/kg K	0.92																				
MECHANICAL PROPERTIES AT ROOM TEMPERATURE																					
Orientation		Longitudinal					Transverse					Short Transverse -									
Thickness	> 25	40	63	90	115	6	12.5	25	40	63	90	115	25	40	63	90	115				
mm	40	63	90	115	140	12.5	25	40	63	90	115	140	40	63	90	115	140				
f_t MPa	A																				
	B	438	438	438	438	418	438	438	428	418	418	408	397	387	387	377	377				
	S	430	430	430	430	410	430	430	420	410	410	400	390	380	380	370	370				
t_2 MPa	A																				
	B	319	319	319	319	319	288	288	288	278	278	278	267	267	267	267	252				
	S	310	310	310	310	310	280	280	280	270	270	270	260	260	260	260	245				
t_1 MPa	A																				
	B	299	299	299	299	303	264	264	263	254	254	256	246	247	247	249	232				
	S	290	290	290	290	294	256	256	257	247	247	249	239	240	240	242	225				
E GPa		74	74	74	74	74	74	74	74	74	74	74	74	74	74	74	74				
ν		0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33				
c_1 MPa	A																				
	B																				
	S																				
c_2 MPa	A																				
	B						309	309	309												
	S						300	300	300												
E_c GPa		75	75	75	75	75	75	75	75	75	75	75	75	75	75	75	75				
f_{50} MPa	A																				
	B	254	254	254	254	243	254	254	248	243	243	237	231	225	225	219	219				
	S	249	249	249	249	238	249	249	244	238	238	232	226	220	220	215	215				
Q_1 MPa	A																				
	B	153	153	153	153	138	138	138	133	133	133	129	129	129	129	129	121				
	S	149	149	149	149	134	134	134	130	130	130	125	125	125	125	118	118				
F_q MPa	A																				
	B	320	320	320	320	305	320	320	313	305	305	298	290	283	283	275	275				
	S	314	314	314	314	299	314	314	307	299	299	292	285	277	277	270	270				
G GPa		28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28				
b_{10} MPa	A																				
	B	473	473	473	473	473	427	427	427	412	412	412	396	396	396	396	373				
	S	459	459	459	459	459	414	414	414	400	400	400	385	385	385	385	363				
f_b MPa	A																				
	B																				
	S																				
Toughness																					
K_{IC} MPa m ^{1/2}	L-T	30.4-38.1-46.1(6)				T-L	29.2-32.1-37.7(3)				S-L										
	L-S					T-S					S-T										

Selecting the material to use

Basically when we select a material we are looking for '**strength**' but strength is a generic word and requires careful definition.

We have already seen that structures require 'strength' in many ways such as: -

Tensile (yield or UTS)

Compressive (yield)

Buckling

Shear (yield or UTS)

also fatigue life, stress corrosion resistance etc. are '**strength**' characteristics.

So, when we select a material we must know the exact purpose for which the material will be used and the type of loadings that will be critical to its failure.

Let us first consider the case of the airframe of a subsonic airliner and determine suitable materials for its construction.

A main design criteria for aircraft is minimum weight. The lighter the aircraft for given dimensions then less lift is needed for take-off and thus less thrust required from the engine thus less fuel is required. Alternatively, if the engines can lift a certain total weight then the less structure mass the larger the payload that can be carried. In terms of an airliner this means more paying passengers and thus a more economically competitive aircraft.

So what we want is some kind of assessment of each candidate material which will tell us if it is **mass efficient** and we do this by the use of **merit indices**.

First let us consider the case of **Rods**.

Consider a pin-jointed rod of cross sectional area A carrying a tensile force F .

One way of assessing the structural efficiency of this part is to compare the applied force (F) with the weight (W) of the component i.e. F/W , that is necessary to carry the force without failure.

Now $F = \sigma A$ (σ = tensile stress)
and $W = \rho AL$ (ρ = rod density)
then :-

$$\frac{F}{W} = \frac{\sigma A}{\rho AL} = \frac{\sigma}{\rho} \times \left(\frac{1}{L} \right)$$

This thus gives a criteria for comparing the structural efficiencies of rods of fixed length L made from different materials namely σ/ρ

If $\sigma_{\max}$ is taken to be the highest tensile stress, which the material can safely carry, then the most effective material is that having the largest value of $\sigma_{\max}/\rho$

This identifies the lightest rod that can withstand the force F along a length L and is known as the **specific strength merit index**.

In some particular cases the strain (ϵ) may be the limiting feature thus: -

$$E = \sigma/\epsilon \quad \text{therefore} \quad \sigma = E\epsilon$$

$$\text{and} \quad F = A\sigma \quad \text{hence} \quad F = AE\epsilon$$

$$\text{Thus} \quad \frac{F}{W} = \frac{AE\epsilon}{\rho AL} = \frac{E}{\rho} \times \left(\frac{\epsilon}{L} \right)$$

Therefore in this case the most effective material would be that with the largest value of E/ρ and is known as **the specific stiffness merit index**.

If the rod is loaded in compression these indices will still be applicable if the rod is thick and can carry stresses upto $\sigma_{\max}$ but if the rod is slender and can fail by buckling then a new index is required.

$$\text{Now } F_e = \frac{n^2 \pi^2 EI}{L^2} \quad \text{where } n \text{ is a constant depending on the type of end fixings.}$$

$$\text{Now the weight of a unit length of rod is } W = \rho \pi r^2$$

$$\text{therefore } r^2 = \frac{W}{\rho \pi} \quad \text{or} \quad r^4 = \frac{W^2}{\rho^2 \pi^2}$$

For the rod $I = \pi r^4/4$ hence $r^4 = \frac{4I}{\pi} = \frac{W^2}{\rho^2 \pi^2}$

Thus $I = \frac{W^2}{4\rho^2 \pi}$ and as $F_e = n^2 \pi^2 EI/L^2$

then EI is a constant as F_e , π^2 and L^2 are constants therefore multiplying by E :-

$$E \left(\frac{W^2}{\rho^2} \right) = 4\pi EI = \text{constant}$$

Therefore $W^2 \propto \frac{\rho^2}{E}$

Therefore $W \propto \frac{\rho}{\sqrt{E}}$

Thus for the lightest rod to support a load F , $\frac{\rho}{\sqrt{E}}$ must be the **lowest** value. To give a higher merit to a lower weight the **specific buckling (rods) merit index** is chosen to be $\frac{\sqrt{E}}{\rho}$

The skin of an aircraft usually consists of thin sheets. If the force carried by the sheet is tensile then the specific strength merit index derived for rods applies. However, if a compressive force is supported then it is usually limited by elastic buckling of the sheet.

Now the thickness of a flat panel can only be varied in one dimension.

Let the second moment of area per unit width of panel = $I = \text{constant} \times t^3$ [ie $bd^3/12$]

Then for n panels $\text{constant} \times t^3 = \frac{FL^2}{\pi^2 nE}$ from the Euler equation

Therefore
$$t^3 = \frac{FL^2}{\pi^2 nE} \times \text{constant}$$

The weight of n panels per unit width = W

$$W = n \rho L$$
 then substituting for t

$$= L n \rho \sqrt[3]{\frac{FL^2}{\pi^2 nE}} \times \text{constant}$$

re- ordering the equation then
$$W = \text{constant} \times n^{2/3} \left(\frac{\rho}{\sqrt[3]{E}} \right) L^{5/3} \sqrt[3]{F}$$

hence
$$F/W = \text{constant} \times \frac{1}{n^{2/3}} \left(\frac{\sqrt[3]{E}}{\rho} \right) \left(\frac{F^{2/3}}{L^{5/3}} \right)$$

now as the maximum load (F) that the panel must carry is a constant for any material then the **specific sheet buckling merit index** is $\frac{\sqrt[3]{E}}{\rho}$

However it is more commonly expressed as E/ρ^3

So, on the basis of merit indices we can evaluate candidate materials for use in the airframe.

Table 1 shows candidate materials for an airliner structure and their merit indices. It can be seen from table 1 that for specific strength CFRP and piano wire score high. For specific stiffness beryllium leads the list with CFRP in second place. For buckling of rod and sheet balsa wood beryllium and CFRP head the table. It might be noted that many model aircraft are made from balsa wood illustrating its

MATERIAL	DENSITY $\rho(\text{kgm}^{-3})$	MAX STRESS $\sigma(\text{MPa})$	YOUNGS MODULUS $E(\text{GPa})$	σ/ρ $\text{MPam}^3\text{kg}^{-1}$	E/ρ $\text{GPam}^3\text{kg}^{-1}$	$E^{1/2}/\rho$ $(\text{GPa})^{1/2}\text{m}^3\text{kg}$ $\times 10^{-3}$	E/ρ^3 $\text{GPam}^9\text{kg}^{-3}$ $\times 10^{-9}$
Carbon fibre reinforced plastic	1500	1050	190	0.7	0.13	9.2	56
Glass fibre reinforced plastic	2000	1240	48	0.6	0.025	3.5	6
Balsa wood	144	9.7	3.3	0.07	0.023	12.6	1100
Piano wire	7800	3000	210	0.39	0.027	1.9	0.44
Aluminium alloy	2810	500	70	0.18	0.025	3.0	3.2
Aluminium	2700	90	70	0.03	0.026	3.1	3.6
Stainless steel	7855	980	185	0.13	0.024	1.73	0.4
Beryllium	1480	272	290	0.18	0.2	11.5	89
Titanium alloy	4420	900	110	0.2	0.025	2.3	1.2
Magnesium alloy	1780	185	45	0.1	0.025	3.8	8

Table 1—Merit indices for various materials

advantages such as low density in those machines. However, merit indices are only a measure of the mass efficiency and other material parameters must be taken into account before selecting the preferred material.

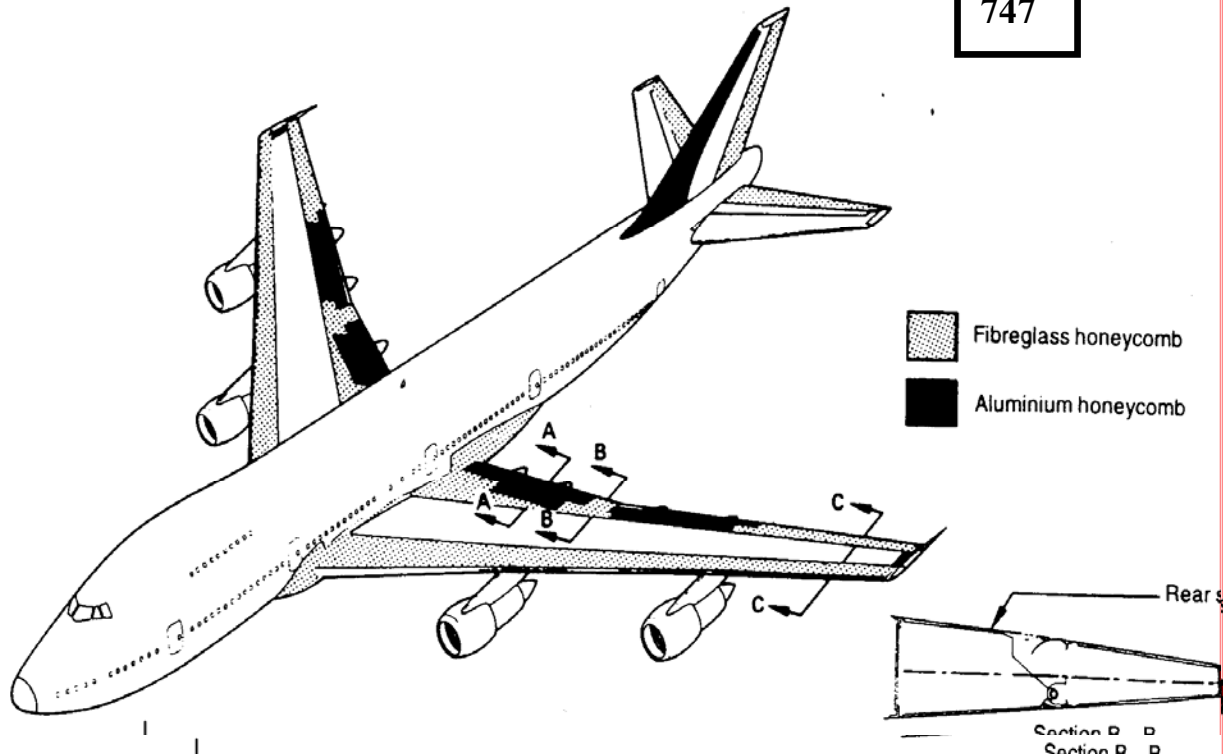
Considering first specific strength, (see table 2) reinforced plastics head the list but are not currently used as the basic airframe of airliners, mainly due to reinforced plastics not yet being considered to be fully proven materials. Fatigue, crack propagation, moisture, vulnerability to damage and other parameters are still being assessed for reinforced plastics. The safety of an airliner has top priority and a fully proven material is essential.

SPECIFIC STRENGTH	SPECIFIC STIFFNESS	BUCKLING (ROD)	BUCKLING (SHEETS)
REINFORCED PLASTICS	BERYLLIUM	BALSA WOOD	BALSA WOOD
PIANO WIRE	CFRP	BERYLLIUM	BERYLLIUM
BERYLLIUM	PIANO WIRE	CFRP	CFRP
ALUMINIUM ALLOY	ALUMINIUM	MAGNESIUM ALLOY	MAGNESIUM ALLOY
TITANIUM ALLOY	ALUMINIUM ALLOY	ALUMINIUM ALLOY	GFRP
	STAINLESS STEEL		ALUMINIUM ALLOY
	TITANIUM ALLOY		
	MAGNESIUM ALLOY		

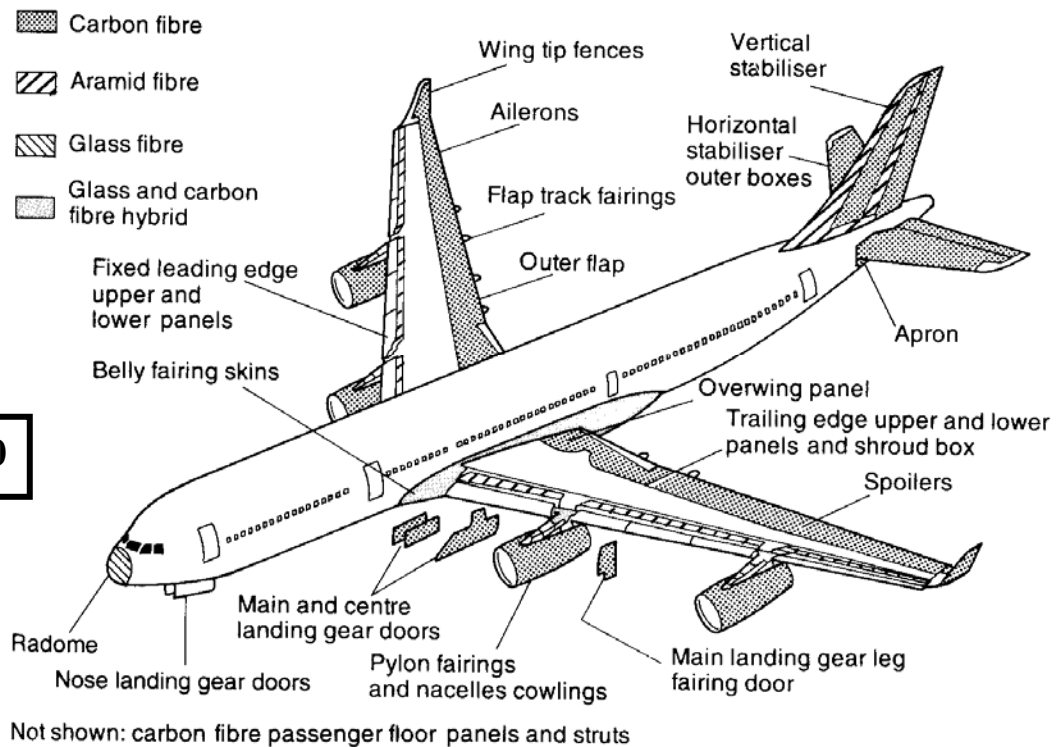
Table 2—Highest merit indices materials

No aircraft manufacturer has yet taken the risk of employing a total reinforced plastic airframe, as failure would almost certainly bankrupt the company due to the loss of the massive investment required in new airliners.

747



A340



It should be noted, however, that gliders and military aircraft primary structure have been constructed of fibre reinforced plastic for sometime, and that both Airbus and Boeing are now using FRP in aircraft primary structures.

Piano wire is next on the list but this is obviously not suitable for an airframe due to their form.

Next come the metals beryllium, al. alloy and titanium alloy.

Beryllium is very costly to process and is also toxic and is thus ruled out in this application.

Titanium alloy has good toughness and resistance to corrosion but aluminium alloy is preferred, due to its lower cost than titanium, and has no major problem areas.

For **Specific Stiffness** : The only new competitors to aluminium alloy are aluminium (ruled out due to its low specific strength) and magnesium (ruled out due to poor corrosion resistance).

For **Buckling**: The only new competitor is balsa wood and this is ruled out due to its low specific strength, bulkiness and susceptibility to damage.

Thus, the material currently used for the airframes of airliners is aluminium alloy. Depending upon the aluminium alloy used, the thin sheets used for the wings and fuselage may be clad with thin layers of pure aluminium to give better corrosion resistance.

Another feature that over rides the merit index value is where space (volume) is at a premium and the highest absolute values of σ and E are sought regardless of the density. This may be the undercarriage of the aircraft, which is required to be stowed in a small place within the wing.

Thus, in selecting materials many parameters must be considered, the basic ones being: -

- . mass and volume efficient for the given load conditions
- . toughness
- . corrosion resistance
- . cost of material, manufacture, and assembly.

Tutorial

Question 1

- (a) Explain briefly what is meant by the term ‘merit index’
- (b) Select from table Q1 the material which has the highest merit index for:-
 - (i) specific strength
 - (ii) specific stiffness
 - (iii) specific buckling (sheet)

Material	Density ρ (kg/m^3)	Strength σ_{uts} (MN/m^2)	Elastic Modulus E (GN/m^2)
Sitka spruce	384	35	9
Aluminium alloy	2800	520	74
Titanium alloy	4500	450	117
Magnesium alloy	1780	185	45
Unidirectional Carbon Fibre Composite (CFRP)	1500	1050	189
Unidirectional Glass Fibre Composite (GRP)	1900	960	38
Kevlar composite	1400	1240	76
Woven glass cloth	1900	408	20

Table Q1—Typical properties of some aircraft structural materials

Question 2

(a) It is required to design an aircraft component with a cylindrical cross-section to a tensile load of 1MN and an Euler buckling load of 0.75MN. The component, which is 1m long between pinned ends, will, in general, be designed by only one of these loads: both criteria will not apply simultaneously. The materials available for the design, together with their relevant properties, are listed below. Which material will provide the lightest component?

Use ultimate load factors of 1.33 in tension and 1.50 in buckling. The outer diameter, D , of the component must not exceed 110mm, and the ratio of outer to inner diameter, d , is fixed at $D = 1.25d$. The effect of end fittings is to be neglected.

(b) What other points should be considered in selecting a material.

Table Q2 -----Material data

Property	Steel	Aluminium Alloy	Titanium Alloy	Beryllium Alloy	Carbon Composite
Ultimate stress (MN/m^2)	1400	560	1050	420	740
Young's modulus (GN/m^2)	210	70	110	315	154
Density (kg/m^3)	7850	2720	4450	1730	1440